

Titanic Dataset — EDA Summary Report

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1. Data Preprocessing Decisions

- Missing values: Cabin was dropped (missing in ~77% of rows — too sparse to impute reliably). Age was filled with its median. Embarked was filled with its mode.
- Duplicates: checked, none required removal.
- Outliers: IQR method applied to Fare. A large number of high-fare passengers (first class) fell outside the IQR bounds; these were capped rather than removed, since dropping ~13% of rows would meaningfully shrink the dataset and these are legitimate high fares, not data errors.
- Categorical encoding: Sex label-encoded (female = 0, male = 1). Embarked one-hot encoded (dropping the first category to avoid redundancy).
- Feature scaling: Age and Fare standardized with StandardScaler (mean = 0, std = 1); Pclass, SibSp, and Parch left as-is since they're already small integer ranges.

2. Key Visualizations

Survival counts by Sex and Pclass:

Age and Fare distributions by survival outcome:

3. Correlation Analysis

Feature Pair	Correlation	Meaning
Sex ↔ Survived	-0.54	Strongest predictor — being male sharply lowers survival odds
Pclass ↔ Survived	-0.34	Lower class (higher Pclass number) reduces survival odds
Fare ↔ Survived	0.32	Higher fare (proxy for class) increases survival odds
Embarked_S ↔ Survived	-0.15	Weak — Southampton passengers slightly less likely to survive
Age ↔ Survived	-0.06	Very weak linear relationship on its own

4. Key Insights

- Sex is the single strongest predictor of survival — women survived at a much higher rate than men (correlation -0.54).
- Passenger class matters: 1st class passengers had notably higher survival rates than 3rd class (Pclass correlation -0.34).
- Fare correlates positively with survival (0.32), largely because it tracks passenger class — wealthier passengers had better access to lifeboats.
- Age has only a weak linear relationship with survival on its own, though very young children showed somewhat higher survival in the raw distributions.
- Cabin data was too sparse to use directly (77% missing) and was dropped rather than imputed, to avoid introducing unreliable values.

5. Machine Learning Readiness Summary

The dataset is preprocessed and ready for baseline modeling:

- No missing values remain; Cabin dropped, Age and Embarked imputed.
- Categorical features (Sex, Embarked) fully numeric.
- Age and Fare scaled; outliers in Fare capped rather than removed, preserving all 891 rows.
- Target class (Survived) is moderately imbalanced (62% died / 38% survived) — worth using stratified train/test split and checking precision/recall, not just accuracy.
- Sex, Pclass, and Fare are the strongest signals and should be prioritized in a baseline model (e.g. Logistic Regression or Decision Tree).